

Vandan Patel

Secaucus, NJ • vrp77@scarletmail.rutgers.edu •

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EDUCATION

Rutgers University–New Brunswick

B.S. in Computer Science, B.S. in Finance, B.A. in Data Science

New Brunswick, NJ

May 2027

- **GPA:** 3.95 | **Honors:** Rutgers Honors College, Dean's List (All Semesters)

TECHNICAL SKILLS

AI & ML: LLM Agents, Prompt Engineering, OpenAI/Anthropic APIs, RAG Pipelines, Vector Search, LangChain, Embeddings

Languages: Python, SQL, JavaScript, TypeScript

Infrastructure & Tools: AWS (EC2, S3), Docker, Git, WebSockets, REST APIs, Cursor, Windsurf, Claude Code

Frameworks & Libraries: FastAPI, Flask, React, Node.js, PostgreSQL, Supabase, SQLAlchemy, dbt-core, Pandas

WORK EXPERIENCE

Rutgers SEBS, Infrastructure & Facilities Planning

New Brunswick, NJ

Intern, Data Analyst (Automation & Infrastructure)

Sep 2025 – Dec 2025

- Eliminated 40+ annual hours of manual verification for 300+ facility nodes by engineering an automated data validation system
- Optimized data workflows to cut licensing costs by 60% by re-architecting the backend to replace legacy premium connectors
- Expedited data verification cycle by 35% by implementing real-time progress tracking and automated stakeholder notifications

Moweb Technologies

Secaucus, NJ

Intern, Software Engineering (Data)

Jul 2025 – Aug 2025

- Mitigated ETL pipeline failures from 20+ weekly incidents to under 1 within 2-week implementation by engineering null-handling logic in dbt-core using COALESCE functions and not_null data tests to prevent downstream breaks
- Reduced query latency 40% by refactoring dbt data models and optimizing Snowflake warehouse performance for client reporting
- Accelerated data reconciliation 30% by engineering LLM-powered quality check automation using prompt engineering, reducing manual validation overhead and improving dataset consistency

PROJECTS

Edgar Agent | *Python, FastAPI, x402, LLMs, Claude MCP*

Dec 2025 – Present

- Architected a pay-per-request API for SEC 10-K filings, returning schema-validated financial data and risk factors via a gated quote/nonce flow with cryptographic signature verification
- Engineered an LLM-powered extraction pipeline to transform unstructured EDGAR HTML into structured financials (revenue, net income, fiscal year-end) with type validation to prevent malformed outputs
- Built an MCP server exposing the service as a tool for autonomous agents, enabling standardized financial data access without manual scraping

Algorithmic Stock Trading Bot | *Python, Alpaca API, Pandas*

Nov 2024 – Aug 2025

- Devised real-time monitoring system processing 50 stocks with 5-point rolling window analysis via WebSockets, implementing momentum detection with configurable thresholds
- Ensured zero downtime for 400+ calls/min by architecting fault-tolerant pipeline with auto-failover and automated risk guardrails

GrindSheet – Social Fitness Analytics PWA | *React, TypeScript, Supabase, TailwindCSS*

May 2025 – Aug 2025

- Developed multi-factor scoring algorithm processing 8 weighted categories for 100+ weekly logs with PR detection and ranking
- Built real-time sync with Supabase WebSockets, reducing query overhead by 80% through optimized filtering and memoization
- Launched production PWA with React 18 and TypeScript featuring offline-first architecture and centralized state management

Sunny – Live Insurance Co-Pilot (Microsoft Hackathon) | *Python, FastAPI, Pathway, LLM*

Jun 2025

- Architected streaming ETL pipeline processing 100+ events from 6 live data sources with sub-2-second latency
- Devised RAG pipeline with vector search, delivering real-time risk alerts from multi-source data for LLM recommendations
- Designed geospatial risk scoring engine with asynchronous data fetching, rate limiting, and failure recovery

OroGenie – Financial Analytics Dashboard | *React, Flask, Yahoo Finance API, SQL*

Jun 2024 – Aug 2024

- Engineered a full-stack portfolio analytics platform using React and Flask, visualizing real-time performance metrics for risk-managed trading strategies
- Implemented Modern Portfolio Theory concepts to calculate and display live risk exposure, asset allocation trends, and volatility metrics for user portfolios
- Established a distributed data pipeline integrating Yahoo Finance and Webull APIs to aggregate market data, enabling users to identify trade improvement opportunities

HACKATHONS AND ENGINEERING LEADERSHIP

Rutgers Shark Tank – "PerkPal"

Piscataway, NJ

Business & Technical Lead, Startup Competition Finalist

Mar 2025

- Led cross-functional team of 3 to top 6/30 finish by conducting competitive analysis and identifying \$2.4B+ market opportunity
- Prototyped live proof-of-concept rewards aggregation platform using React and Selenium, presenting scalable technical architecture and go-to-market strategy to investor panel

ACTIVITIES & INTERESTS

Activities: Quantitative Finance Club, Rutgers Mobile App Development Club, RU Sangam, Venture Capital Club

Interests: Mountain Biking, Calisthenics, Powerlifting, Options Trading, Automotive Enthusiast